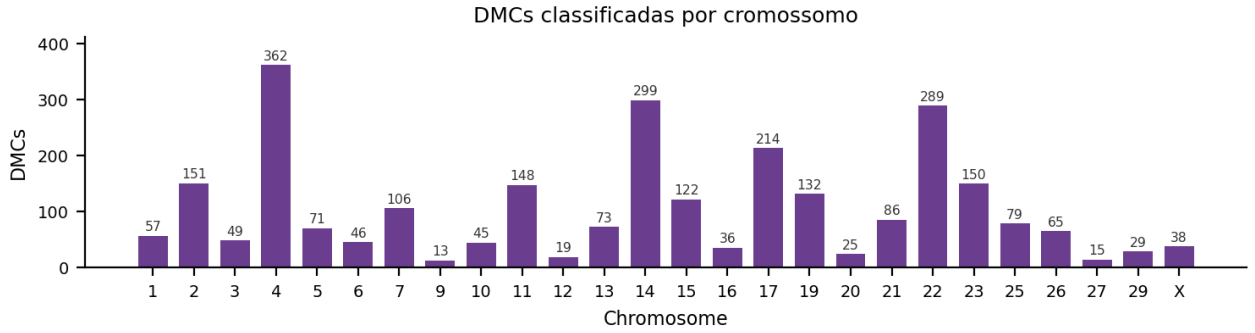
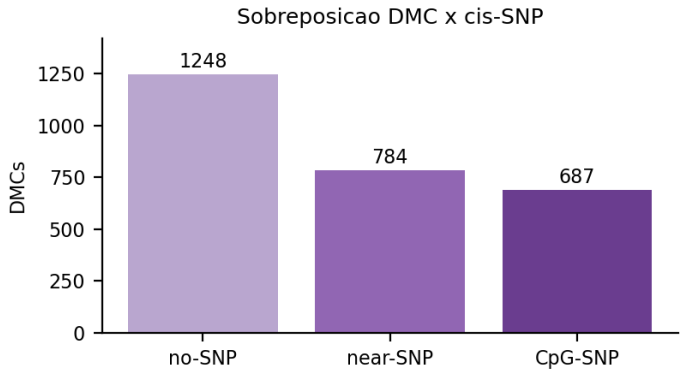
cis-SNPs over differentially methylated cytosines

Contraste entre subespecies de *Bos taurus* · painel de 108 genes · ARS-UCD1.2 · 1000 Bull Genomes Run 8

2.719	25.3%	45.9%	80	65 / 15
DMCs classificadas	em SNP CpG-disruptivo	sem SNP em +-10 bp	SNPs quebra-motivo	SP1 / KLF4





#	SNP (chr:pos)	Gene	Motivo (TF)	Proximidade	Feature
1	25:34348773	HSPB1	SP1	tight(<=10)	downstream distal
2	10:76687664	HSPA2	SP1	tight(<=10)	downstream distal
3	16:51460931	ISG15	SP1	tight(<=10)	downstream distal
4	16:51461454	ISG15	KLF4	tight(<=10)	downstream distal
5	25:34348786	HSPB1	SP1	loose(<=25)	downstream distal
6	16:51461343	ISG15	SP1	loose(<=25)	downstream distal
7	14:639233	HSF1	SP1	tight(<=10)	upstream distal
8	14:642016	HSF1	SP1	tight(<=10)	upstream distal
9	21:66950681	HSP90AA1	SP1	tight(<=10)	upstream distal
10	15:33668789	HSPA8	KLF4	tight(<=10)	downstream distal
11	15:33683541	HSPA8	KLF4	tight(<=10)	promoter
12	3:92330755	HSPB11	SP1	tight(<=10)	gene body
13	17:69268527	LIF	SP1	tight(<=10)	gene body
14	17:69274702	LIF	KLF4	tight(<=10)	promoter
15	15:75919079	MAPK8IP1	SP1	tight(<=10)	promoter
16	15:75937201	MAPK8IP1	SP1	tight(<=10)	gene body
17	14:594484	DGAT1	SP1	tight(<=10)	upstream distal
18	4:68853834	HOXA10	SP1	tight(<=10)	upstream distal
19	14:633663	HSF1	SP1	tight(<=10)	promoter
20	14:639232	HSF1	SP1	tight(<=10)	upstream distal
21	17:69278517	LIF	SP1	tight(<=10)	upstream distal
22	17:69278542	LIF	SP1	tight(<=10)	upstream distal
23	15:75936196	MAPK8IP1	KLF4	tight(<=10)	gene body
24	14:642174	HSF1	SP1	tight(<=10)	upstream distal
25	13:51461167	HSPA12B	SP1	tight(<=10)	promoter
26	17:69268529	LIF	SP1	tight(<=10)	gene body
27	14:594235	DGAT1	SP1	tight(<=10)	upstream distal
28	4:68877111	HOXA10	SP1	tight(<=10)	downstream distal

#	SNP (chr:pos)	Gene	Motivo (TF)	Proximidade	Feature
29	23:27521893	HSPA1A	KLF4	tight(<=10)	promoter
30	2:6275088	MSTN	SP1	tight(<=10)	upstream distal
31	19:33275408	STAT3	KLF4	tight(<=10)	gene body
32	4:68880549	HOXA10	SP1	tight(<=10)	downstream distal
33	14:642012	HSF1	SP1	tight(<=10)	upstream distal
34	17:69257297	LIF	SP1	tight(<=10)	downstream distal
35	15:75918757	MAPK8IP1	KLF4	tight(<=10)	promoter
36	19:33275407	STAT3	KLF4	tight(<=10)	gene body
37	14:642167	HSF1	SP1	tight(<=10)	upstream distal
38	14:642010	HSF1	SP1	tight(<=10)	upstream distal
39	13:61270927	BCL2L1	SP1	tight(<=10)	promoter
40	4:68859002	HOXA10	SP1	tight(<=10)	upstream distal
41	4:68859385	HOXA10	SP1	tight(<=10)	upstream distal
42	14:641234	HSF1	SP1	tight(<=10)	upstream distal
43	23:27521903	HSPA1A	SP1	tight(<=10)	promoter
44	13:61270928	BCL2L1	SP1	tight(<=10)	promoter
45	11:73971384	DNMT3A	SP1	tight(<=10)	promoter
46	13:51461161	HSPA12B	SP1	tight(<=10)	promoter
47	4:68859004	HOXA10	SP1	tight(<=10)	upstream distal
48	14:594493	DGAT1	SP1	tight(<=10)	upstream distal
49	14:594221	DGAT1	SP1	loose(<=25)	upstream distal
50	19:33275422	STAT3	SP1	loose(<=25)	gene body
51	4:68858983	HOXA10	KLF4	loose(<=25)	upstream distal
52	13:19971633	ITGB1	SP1	loose(<=25)	promoter
53	13:61270933	BCL2L1	SP1	loose(<=25)	promoter
54	14:593636	DGAT1	SP1	loose(<=25)	upstream distal
55	21:66944554	HSP90AA1	SP1	loose(<=25)	promoter
56	15:75934971	MAPK8IP1	SP1	loose(<=25)	gene body
57	15:75938230	MAPK8IP1	KLF4	loose(<=25)	downstream distal
58	13:61270935	BCL2L1	SP1	loose(<=25)	promoter
59	23:17763077	HSP90AB1	KLF4	loose(<=25)	downstream distal
60	23:27521881	HSPA1A	KLF4	loose(<=25)	promoter
61	19:48271454	ERN1	KLF4	loose(<=25)	downstream distal
62	3:15438128	MUC1	KLF4	loose(<=25)	downstream distal
63	27:33319025	STAR	SP1	loose(<=25)	gene body
64	15:51218207	AURKA	SP1	loose(<=25)	upstream distal
65	19:48350837	ERN1	SP1	loose(<=25)	promoter
66	14:639251	HSF1	SP1	loose(<=25)	upstream distal
67	4:68853814	HOXA10	SP1	loose(<=25)	upstream distal
68	26:37146493	HSPA12A	SP1	loose(<=25)	gene body
69	3:92330774	HSPB11	SP1	loose(<=25)	gene body
70	6:36532308	ABCG2	SP1	loose(<=25)	gene body
71	13:62143481	DNMT3B	SP1	loose(<=25)	promoter
72	19:48269549	ERN1	SP1	loose(<=25)	downstream distal
73	5:109514104	LGALS1	SP1	loose(<=25)	promoter
74	27:33331208	STAR	SP1	loose(<=25)	upstream distal
75	6:36475339	ABCG2	SP1	loose(<=25)	promoter
76	14:594506	DGAT1	SP1	loose(<=25)	upstream distal
77	19:48269551	ERN1	SP1	loose(<=25)	downstream distal
78	15:75936623	MAPK8IP1	SP1	loose(<=25)	gene body
79	13:61270945	BCL2L1	SP1	loose(<=25)	promoter
80	4:68880520	HOXA10	SP1	loose(<=25)	downstream distal

cis-SNPs overlapping subspecies DMCs. Among 2.719 differentially methylated cytosines (DMCs) between *Bos taurus taurus* and *Bos taurus indicus* over the 108-gene panel, each DMC was cross-referenced against 1000 Bull Genomes Run 8 variants within ±10 bp: 45.9% carry no SNP, 28.8% sit near a SNP and 25.3% fall on a CpG-disrupting SNP. Bar charts show the chromosomal distribution and the three overlap classes. The table lists the 80 DMCs whose underlying SNP is predicted (motifbreakR) to disrupt an SP1 or KLF4 motif (65 SP1, 15 KLF4). Coordinates on ARS-UCD1.2; SNPs identified by chromosome:position (no rsID/allele by design — SNPs enter as an annotation layer over methylation). Full tables: TableS5_cissnp.tsv (all 2.719 DMCs) and TableS8_motif_dmc.tsv (80 motif-breaking SNPs).

Data: 1000 Bull Genomes Run 8 (ARS-UCD1.2); methylation contrast over the 108-gene panel; motif disruption predicted with motifbreakR.